LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART II

SESSION 7

National Accounts, Productivity & Welfare

From markets, to macro, to decisions that matter — in Nigeria and the world.



Reading: N. Gregory Mankiw, Macroeconomics · Facilitator: Bongo Adi

RECAP

Session 6 in Three Ideas

↗

AD vs AS

Demand shocks move prices and output together; supply shocks move them apart.

±

Measures

Headline for the cost of living; core and trimmed mean for the underlying trend.

✦

Expectations

Inflation is sticky because people price in what they expect.

TODAY'S PLAN

Session 7 at a Glance — 75 Minutes

How we measure the size of an economy — and why the headline number can mislead.



Hook

The day Nigeria's GDP jumped

Measuring GDP

Three approaches; GDP vs GNI; nominal vs real; rebasing

Growth & Living Standards

Oil vs non-oil, per person, PPP, informality

Lab & Discussion

Is Nigeria getting richer?

BY THE END OF TODAY

Session 7 Learning Objectives

You will be able to:

01

Read the National Accounts

Explain GDP, GDI and GNI, nominal versus real, and what a rebasing does — and does not — change.

02

Find the Growth Drivers

Separate oil from non-oil growth and spot which sectors are gaining share.

03

Judge Living Standards

Use per-person, PPP and 'beyond GDP' measures to size a market honestly.

THE HOOK

Bigger Overnight — Richer Overnight?

✦

“Nigeria is now Africa's largest economy.” — headlines, April 2014

In 2014 the statistics office rebased GDP to 2010 prices and the measured economy grew by almost 90% in a single announcement. What changed?

What was counted

Telecoms, Nollywood, e-commerce and other new activities that the old 1990 base barely measured.

What was not changed

No household earned an extra naira that morning. Output was the same; the measurement improved.

What moved with it

Debt-to-GDP and tax-to-GDP ratios fell instantly — the denominator was bigger.

NIGERIAN BUSINESS CASE

Is Nigeria Getting Richer?

The situation

A consumer-goods company plans its five-year strategy. The CEO's slide says 'Nigeria's economy has grown 50% since 2010'. The CFO is less sure the average customer is better off.

The decision question

How much has the economy **really** grown, which sectors are driving it, and has income **per person** risen enough to grow our market?

The data we have

ng_gdp_sectors.csv — real GDP by ten sectors (₦ tn, 2010 prices), nominal GDP on old and new bases, population, 2010–2025; peer_income.csv — ten countries. *Stylised teaching data.*

What a good answer looks like

A one-paragraph view for the board on **real growth, growth per person, and the sectors** that matter for our demand.

PART

01

Measuring the Economy

One number, three ways to get it.

Production, Spending and Income Give the Same Total

Output (production)

Add the **value added** of every sector: farm output, factories, banks, telecoms — minus the inputs they used. This is how NBS builds most of Nigeria's GDP.

Expenditure

Add what is spent on final goods: **C + I + G + (X − M)** — consumption, investment, government and net exports.

Income (GDI)

Add what is earned: wages, profits, rent and interest. In principle equal to GDP; in practice a 'statistical discrepancy' remains.

*Value added avoids double counting: count the bread, not the flour **and** the bread.*

Which 'National Income' Number Answers Which Question?

Measure	What it counts	Why a manager cares
GDP	Value produced **inside** Nigeria, by anyone	Size of the domestic market and activity
GNI	GDP + income earned abroad by Nigerians − income paid to foreigners	Nigeria's GNI is below GDP: profits and interest flow out to foreign owners and lenders
Disposable income	Income households can spend after taxes and transfers (incl. remittances)	Closest to what customers can actually spend
GDP per person	GDP ÷ population	Average living standard — the number that matters for most consumer markets

Remittances from Nigerians abroad do not add to GDP, but they do raise what households can spend.

Taking Prices Out of GDP

%

IN PLAIN ENGLISH

Nominal GDP values output at today's prices; **real** GDP values it at a fixed base year's prices, so it moves only when **quantities** move. The ratio of the two is the **GDP deflator** — the broadest price index we have.

EXAMPLE

In this data nominal GDP rose **6.8×** from 2010 to 2025, real GDP only **1.51×**. Most of the 'growth' in naira was inflation; the deflator rose about **30%** in 2024 alone.

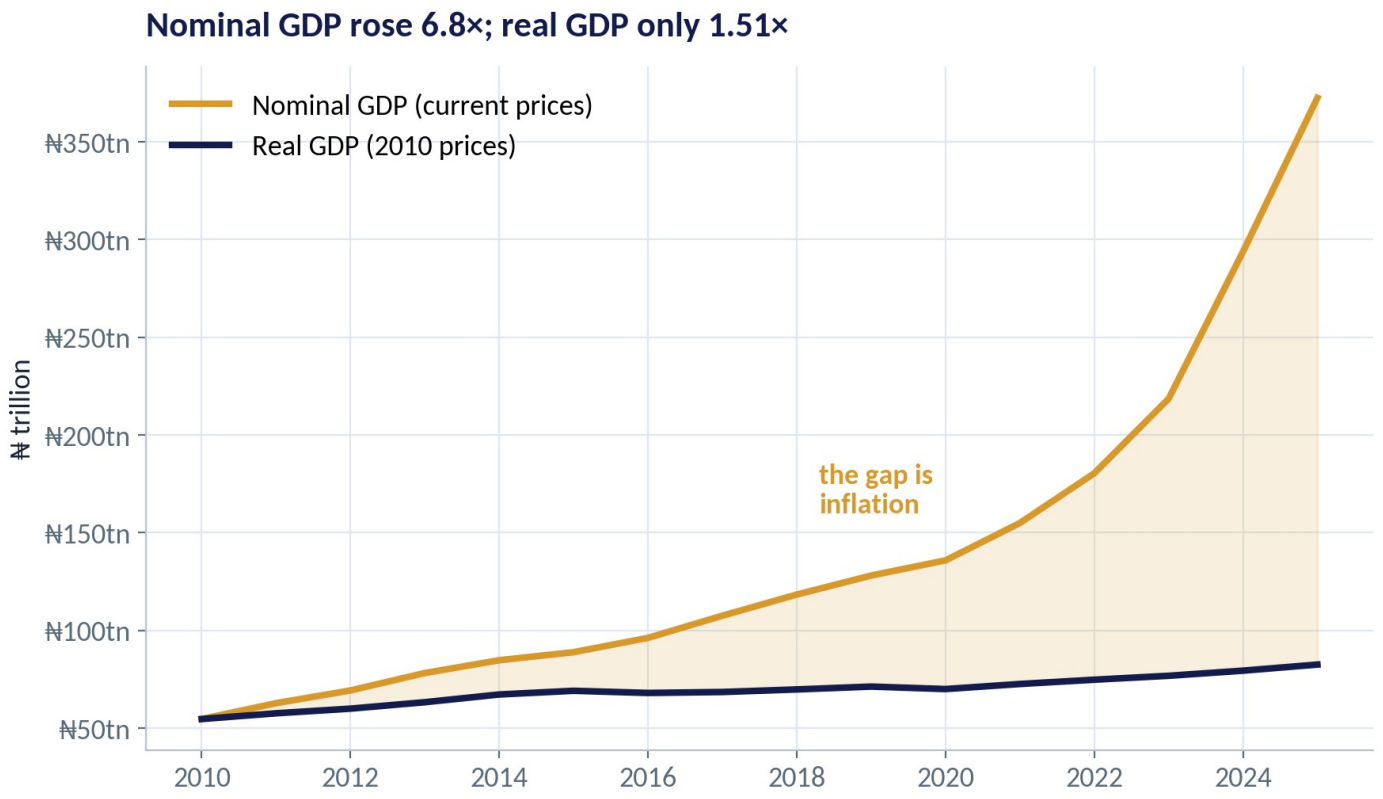
THE FORMULA

$$\text{deflator} = \text{nominal GDP} \div \text{real GDP} \times 100$$

WHY A MANAGER CARES

A market that grows 25% in naira while prices rise 30% is shrinking in volume. Always ask for real growth before celebrating.

Nominal vs Real GDP



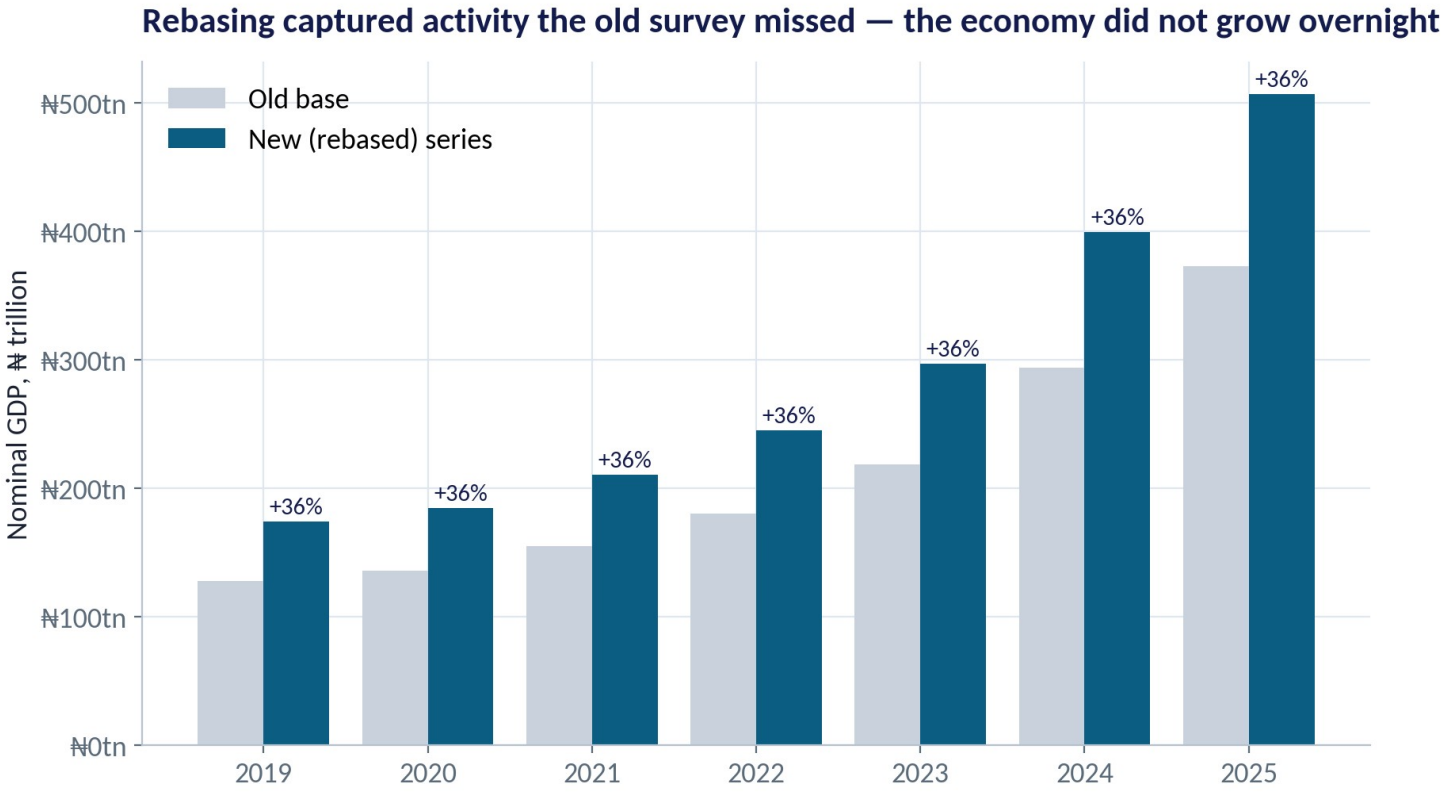
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F04 · replicate: s07_figures.R / .py

WHAT TO SEE

- The gold line is GDP in current naira; the navy line removes inflation.
- From 2010 to 2025 nominal GDP rose **6.8×**; real GDP rose **1.51×**.
- The widening gap after 2022 reflects the inflation surge you saw in Session 6.
- Plan capacity on **real** growth; plan cash and prices on **nominal**.

Same Economy, Bigger Measured GDP



Source: EEGB stylised teaching data. The size of the uplift is illustrative, not the official NBS revision.

Fig S07-F05 · replicate: s07_figures.R / .py

WHAT TO SEE

- Rebasing updates the **base year**, the **sector weights** and the **coverage** of new activities.
- Here the rebased series is about **36%** higher in every year — an illustrative uplift.
- Growth **rates** change much less than **levels**.
- Ratios to GDP (debt, tax, credit) all fall at once — read them with care.

CONCEPT CHECK

Quick Check: What Would You Say?

?

After a rebasing, Nigeria's measured GDP rises 36% while the debt stock is unchanged. Which statement is right?

A. The government can now borrow 36% more safely

B. Debt-to-GDP falls, but the government's ability to pay — its revenue — has not changed

C. Living standards rose 36%

D. Real growth last year was 36%

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

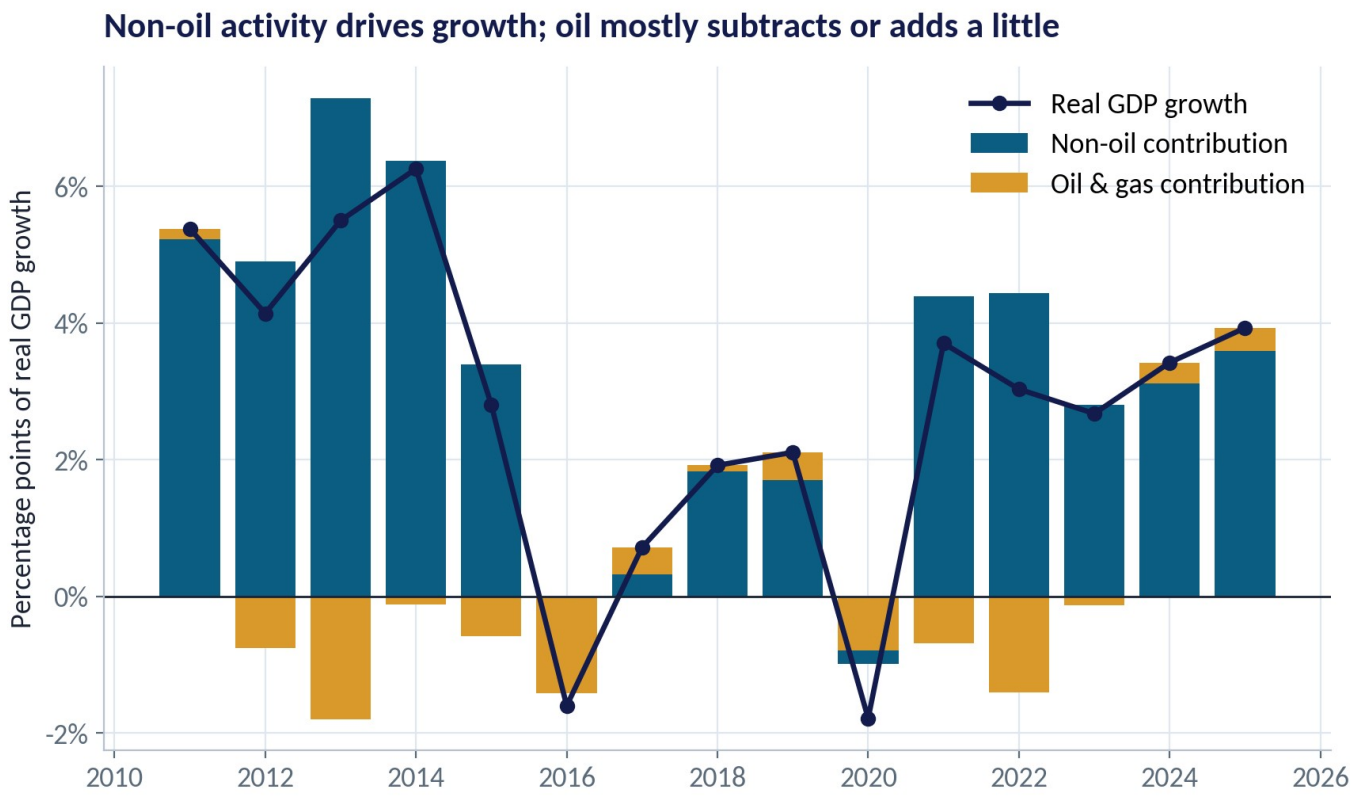
PART

02

Where Growth Comes From

Oil, non-oil, and sectors on the move.

Who Contributes to Growth?



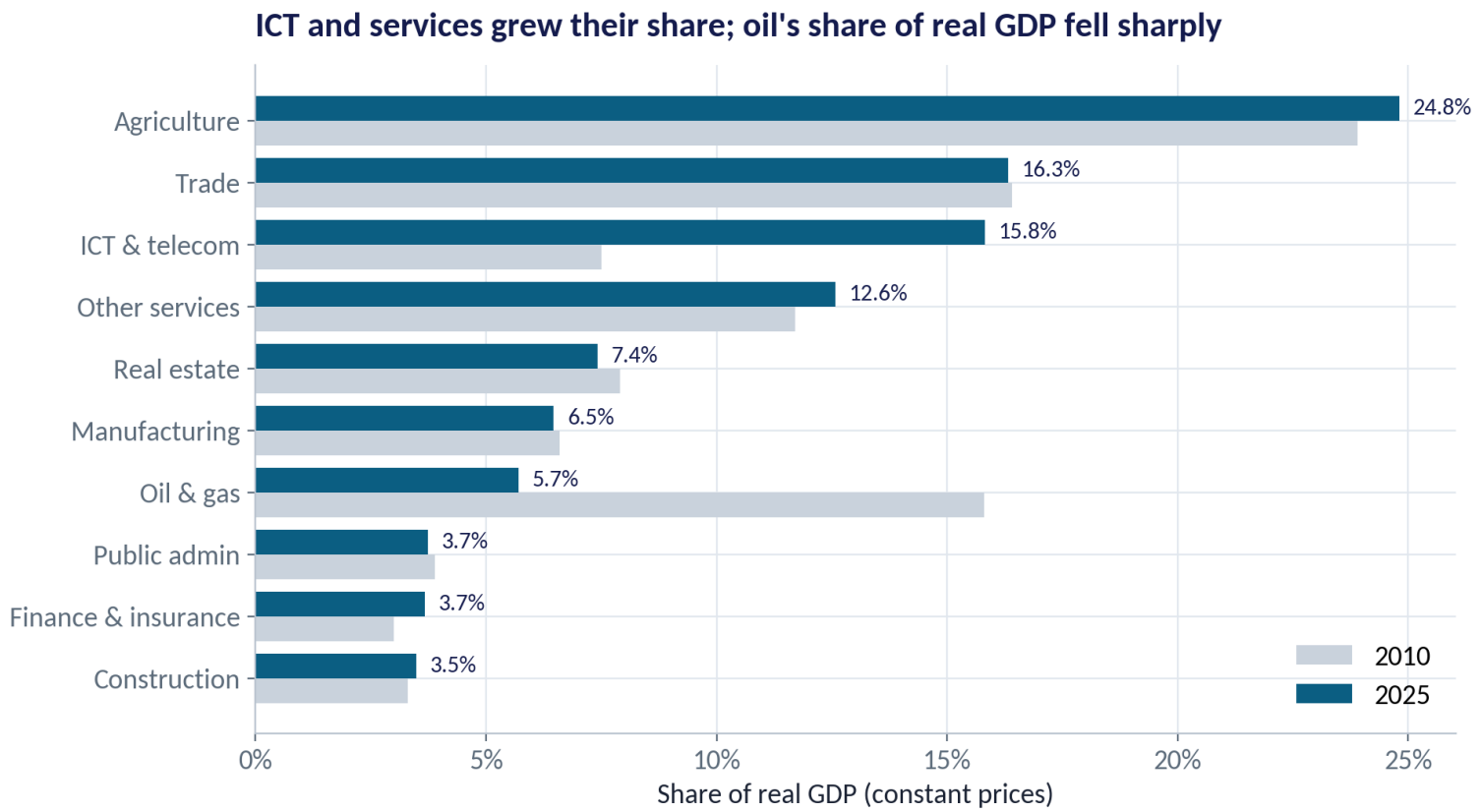
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F01 · replicate: s07_figures.R / .py

WHAT TO SEE

- Bars split each year's growth into **non-oil** (teal) and **oil & gas** (gold).
- Non-oil activity provided almost all the growth; oil subtracted in most years from 2012 to 2023.
- In 2022 oil output fell about **19%** in this data while non-oil grew **4.8%**.
- Oil is small in GDP but huge for **dollars and government revenue** — that is why it still moves the naira.

The Changing Shape of the Economy



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F02 · replicate: s07_figures.R / .py

WHAT TO SEE

- **ICT & telecom** roughly doubled its share, from **7.5%** to **15.8%** of real GDP.
- **Oil & gas** fell from **15.8%** to **5.7%**.
- **Agriculture** and **trade** remain the largest sectors, at about a quarter and a sixth of output.
- Manufacturing stayed near **6.5%** — little sign of industrialisation.

WORKED EXAMPLE

Average Growth by Sector, 2010–2025

Sector	2010 (₦tn)	2025 (₦tn)	Growth/yr
ICT & telecom	4.1	13.0	8.0%
Finance & insurance	1.6	3.0	4.2%
Agriculture	13.0	20.5	3.0%
Manufacturing	3.6	5.3	2.6%
Oil & gas	8.6	4.7	–4.0%

STEP BY STEP

1. Take the **real** (constant-price) value at the start and the end.
2. Divide end by start, then take the **15th root** (15 years).
3. Subtract 1: **(13.0 ÷ 4.1)^(1/15) − 1 ≈ 8.0%** a year for ICT.
4. Compare with population growth of about **2.6%** a year.

A sector growing faster than population is a growing market **per customer**. Only ICT and finance clearly beat population growth here.

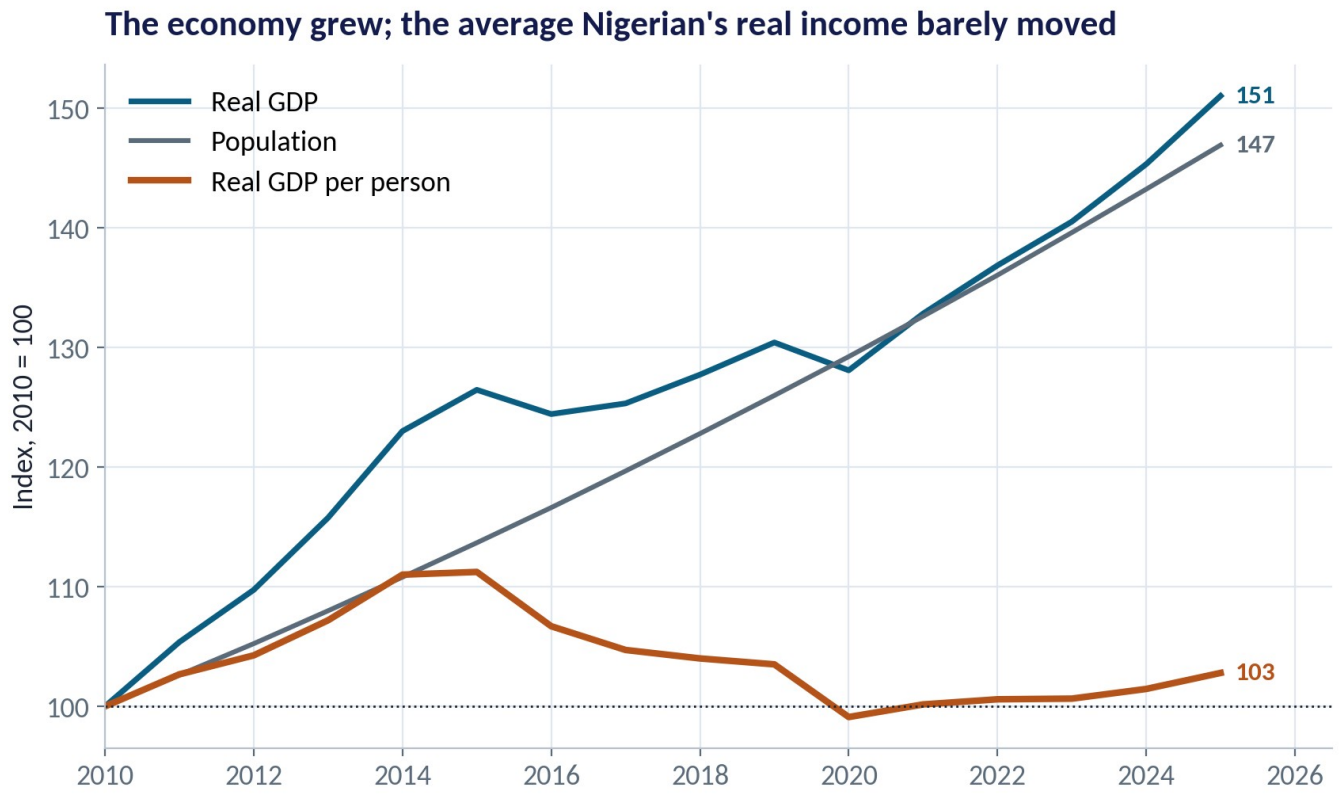
PART

03

Growth and Living Standards

Per person, in purchasing power, and beyond GDP.

Bigger Economy, Same Average Income



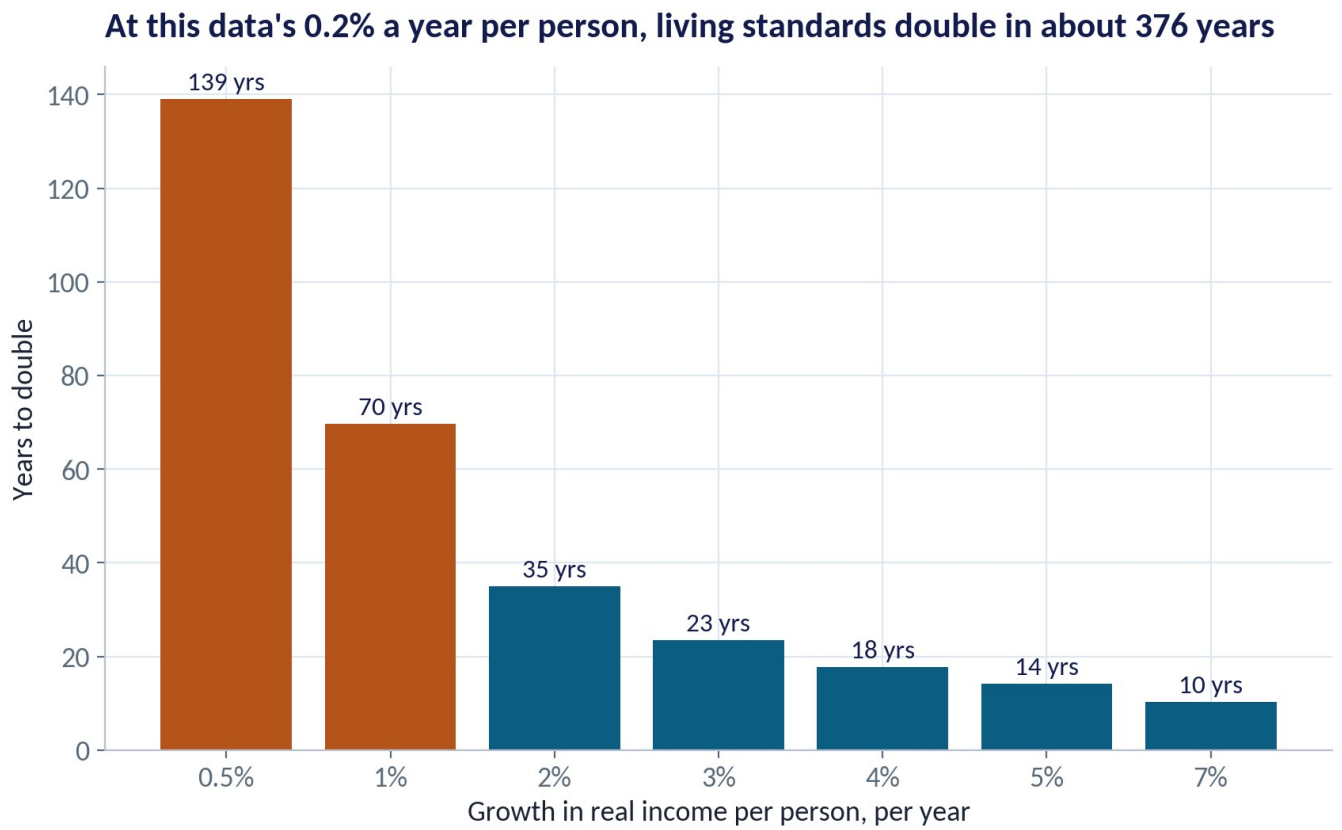
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F03 · replicate: s07_figures.R / .py

WHAT TO SEE

- Real GDP rose to **151** (2010 = 100); population rose to **147**.
- Real GDP **per person** ended at only **103** — almost no gain in fifteen years.
- Population grows about **2.6%** a year; GDP must beat that just to stand still.
- For consumer businesses: more customers, but not richer ones. Volume and value strategies differ.

How Long to Double Living Standards?



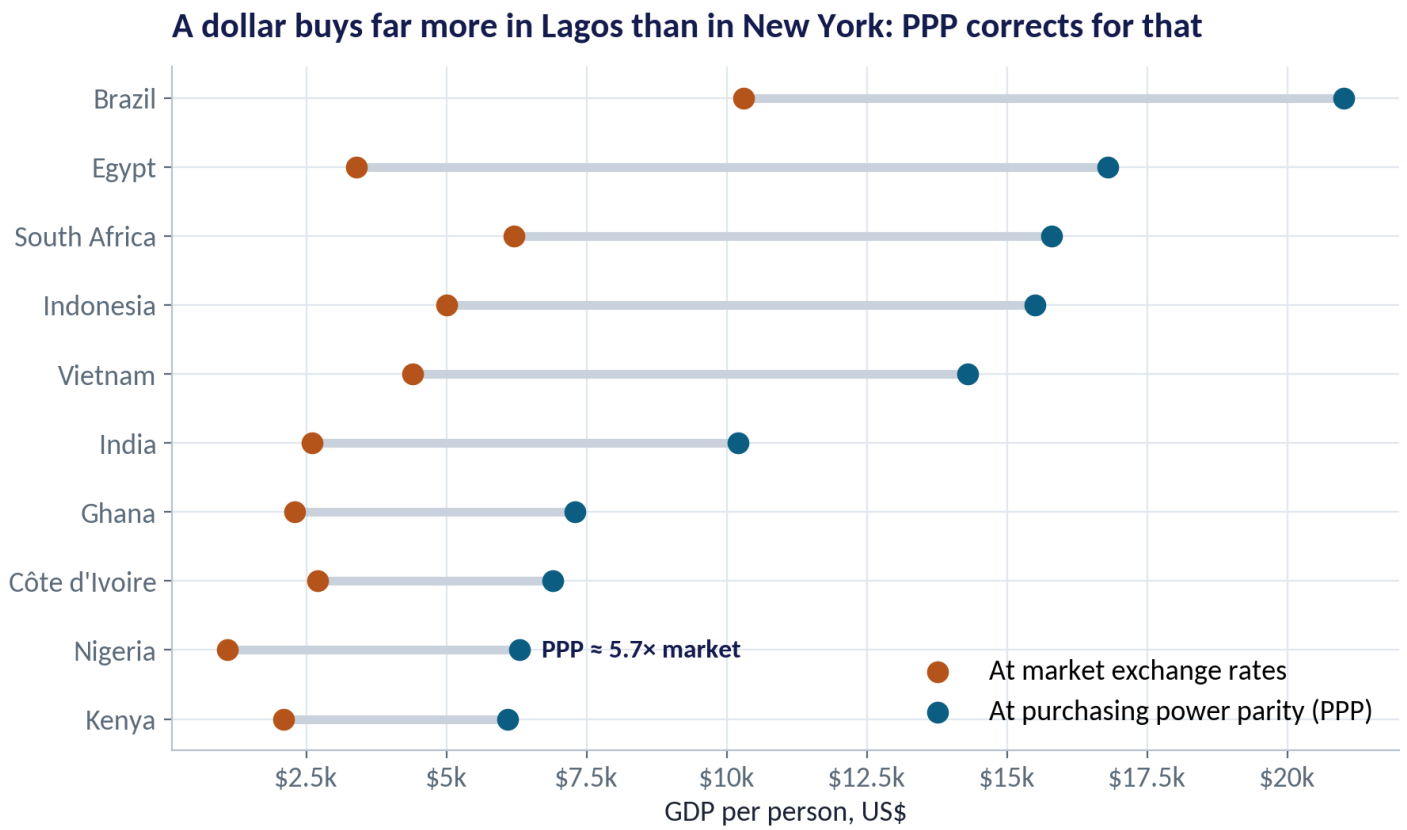
Years to double = ln 2 ÷ ln(1 + g) ≈ 70 ÷ g.

Fig S07-F08 · replicate: s07_figures.R / .py

WHAT TO SEE

- Years to double ≈ **70 ÷ growth rate**.
- At 3% per person, income doubles in about **23 years**; at 7%, about **10**.
- At this data's **0.2%** per person, it takes about **376 years**.
- Small differences in growth, compounded, decide whether a generation gets richer.

Market Rates vs PPP



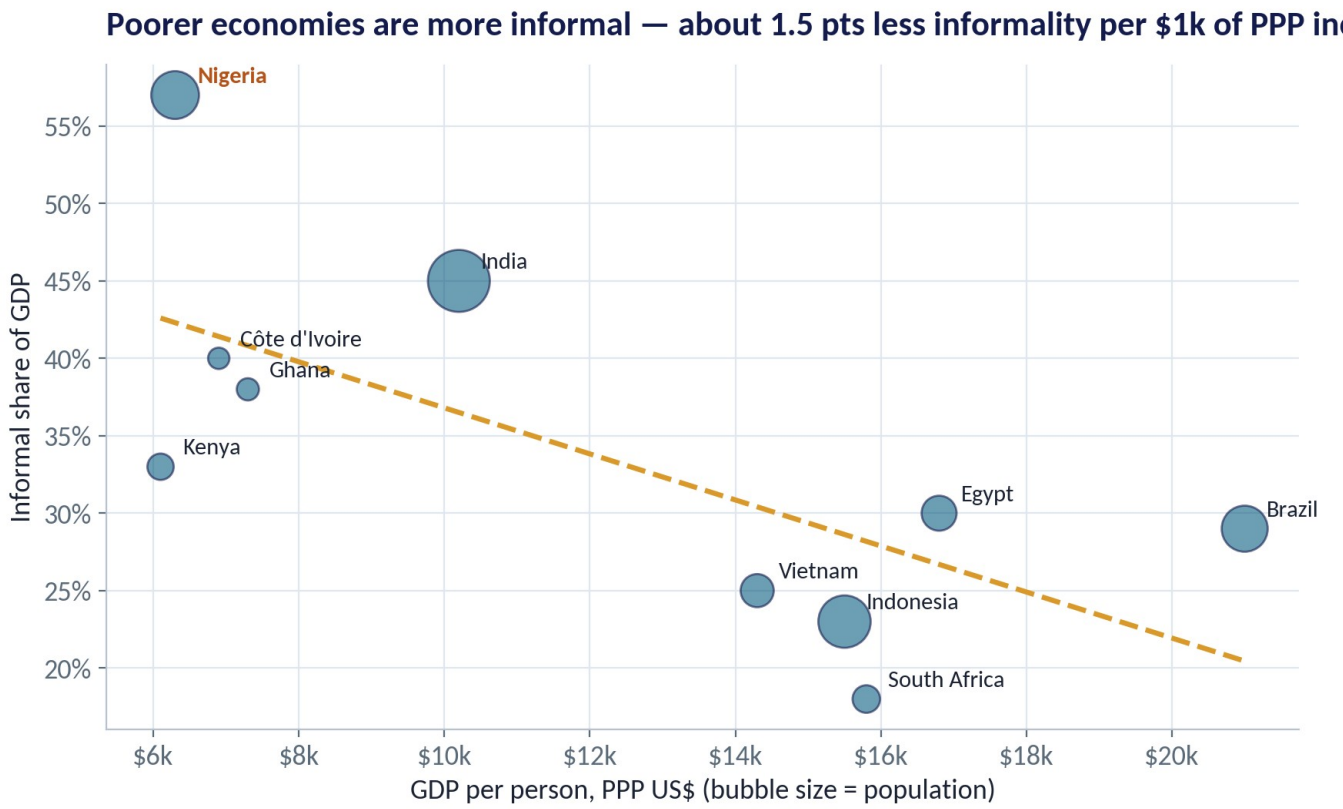
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F06 · replicate: s07_figures.R / .py

WHAT TO SEE

- At market exchange rates, GDP per person here is about **$1,100**; at PPP about **$6,300**.
- PPP adjusts for the fact that haircuts, food and rent cost less in Lagos than in London.
- Nigeria's PPP-to-market ratio (**5.7×**) is the highest in this group — partly because the naira fell so far.
- Use **PPP** to compare living standards; use **market rates** to price imports or report in dollars.

The Economy the Statistics Struggle to See



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S07-F07 · replicate: s07_figures.R / .py

WHAT TO SEE

- Informal firms are unregistered and often unseen by surveys and tax.
- In this data Nigeria's informal share is about **57%** of GDP — the highest of the group.
- Across countries, richer economies are less informal: about **1.5 points** less per $1k of PPP income.
- For managers: much of your market is cash-based, informal and hard to size — triangulate with your own sales data.

What GDP Leaves Out



Distribution

Average income can rise while most households fall behind. Look at poverty and median income.



Unpaid work

Care and household work are not counted, though they sustain the economy.



Depletion

Selling oil counts as income; running down the reserve is not subtracted.



Health & education

The Human Development Index adds life expectancy and schooling to income.

Common Mistakes With National Accounts

THE TRAP	THE BETTER HABIT
✗ “GDP grew 50%, so our market grew 50%.”	✓ Check growth **per person** and in **your customers'** income segment.
✗ Reading nominal GDP growth as real growth.	✓ Deflate: real growth ≈ nominal growth − deflator inflation.
✗ Treating a rebasing jump as new growth.	✓ Rebasing changes the **level**, not what happened last year.
✗ Comparing countries at market exchange rates to judge living standards.	✓ Use **PPP** for living standards; market rates for dollar costs.
✗ Ignoring the informal economy because it is not in the data.	✓ **Triangulate** with your own sales, distributor and payments data.

Quick Check: What Would You Say?

?

Real GDP grows 3.4% and population grows 2.6%. Roughly how fast is real income per person growing, and how long until it doubles?

A. 3.4% a year; about 20 years

B. About 0.8% a year; about 90 years

C. 6.0% a year; about 12 years

D. It cannot be calculated without PPP data

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Is Nigeria Getting Richer?

YOU WILL NEED

Data

`data/ng_gdp_sectors.csv`
`data/peer_income.csv`

R script

`sessions/s07/R/s07_lab.R`

Python script

`sessions/s07/python/s07_lab.py`

⏱ 20 minutes

1. Compute **real GDP growth** year by year.
2. Split growth into **oil and non-oil**.
3. Rank sectors by **average annual growth**, 2010–2025.
4. Compute real GDP **per person** and the years it would take to double.
5. Compare **market-rate and PPP** income across peers.

Deliverable: a short paragraph for the board: real growth, growth per person, and the two sectors that matter most for your demand.

WORDS TO OWN

Today's Vocabulary in Plain English

GDP	Value of all final goods and services produced in a country in a period.
Value added	Output minus the inputs bought from other firms.
GNI	GDP plus net income received from abroad.
Real GDP	GDP at constant prices: shows changes in volume.
GDP deflator	Nominal ÷ real GDP × 100: the broadest price index.
Rebasing	Updating the base year, weights and coverage of the accounts.
PPP	Exchange rate that equalises what money can buy across countries.
Informal economy	Activity that is unregistered and largely unmeasured.
Productivity	Output per worker or per hour; the root of living standards.
Rule of 70	Years to double ≈ 70 ÷ the annual growth rate.

KEY TAKEAWAYS

Session 7 in 5 Ideas

01 GDP can be measured by **output, spending or income**; Nigeria's GNI is below its GDP.

02 Always separate **nominal** from **real**; the deflator tells you how much was inflation.

03 **Rebasing** changes levels and ratios, not what the economy did.

04 **Non-oil** sectors drive growth, but oil still drives dollars and revenue.

05 What matters for customers is income **per person** — and here it has barely moved.



NEXT › Session 8: Monetary Policy

Before next session: Read the HBS Note on Money and Monetary Policy (9-797-076) and the CBN brief on liquidity management. Note how your company's borrowing rate changed between 2023 and 2025.